# 11-1 Modeling

## Point!

### Modeling

(1) ([1]**Model**) : A simplified representation of a target phenomenon.

(2) ([2]**Modeling**) : The process of creating models for events or phenomena.

(3) Classification of Models by Characteristics

[1] ([3]**Static model**) : A model that does not undergo changes over time.
<Examples> Room layout diagrams, the relationship between the radius and volume of a sphere, etc.

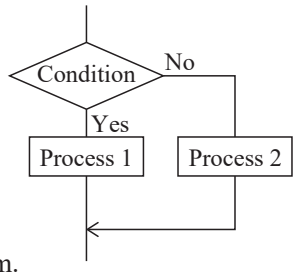
[2] ([4]**Dynamic model**) : A model that is influenced by changes over time.

- ([5]**Deterministic model**) : A model that does not exhibit irregular behavior. There is only one result.
<Examples> Bank account balance, relationship between time and travel distance for a car moving at a constant speed, etc.
- ([6]**Probabilistic model**) : A model that includes irregular behavior. The result is not a single outcome.
<Examples> Outcomes of dice rolls, lottery draws, weather forecasts, etc.

(4) Classification of Models Based on Type of Expression

[1] ([7]**Physical model** (**scale model**)) : A physical representation of an object.

- ([8]**Enlarged model**) : A model that is larger than the actual object.
<Examples> Molecular model, DNA model, etc.
- ([9]**Full-scale model**) : A model that is the same size as the actual object.
<Examples> Model room, life-size dummy, etc.
- ([10]**Reduced model**) : A model smaller than the actual object.
<Examples> Globe, plastic model, etc.



**Flowchart**

[2] ([11]**Diagrammatic model**) : A representation of something using a diagram.
<Examples> Flowcharts, route maps, etc.

[3] ([12]**Mathematical model**) : A representation of something using mathematical formulas or logical expressions.
<Examples> Speed × Time = Distance,
Current × Resistance = Voltage, etc.

(5) Modeling steps

[1] Clearly define the ([13]**purpose**) of modeling.

[2] Clarify the ([14]**elements that constitute the model**) and their relationships.

[3] Determine the ([15]**representation method**) of the model.

## Warm Up

Answer the following questions.

(1) Choose the classification of the models most related to sentences [1] to [4] from the options **A** to **C** below, and answer using the letters.

[1] The relationship between the radius and area of a circle

[2] Outcome of a dice roll

[3] Weather forecast

[4] The relationship between time and distance traveled for a bicycle at a constant speed

**A** Static model **B** Deterministic model **C** Probabilistic model

(2) Choose the classification of the models most related to sentences [1] to [4] from the options **A** to **C** below, and answer using the letters.

[1] Molecular model [2] Newton's equation of motion

[3] Flowchart of a program [4] Plastic model

**A** Physical model **B** Diagrammatic model **C** Mathematical model

### Explanation

(1) [1] **A** [2] **C** [3] **C** [4] **B**

(2) [1] **A** [2] **C** [3] **B** [4] **A**

## Try

Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [5] with the appropriate terms.

> ( [1] ) are referred to as a simplified representation of a target phenomenon. Types of ([1]) include: a ( [2] ) model, which is a physical representation of an object; a ( [3] ) model, which uses mathematical expressions or logical formulas to represent the behavior of phenomena; and a ( [4] ) model, which uses diagrams to represent phenomena. Examples of ([2]) models include full-scale models, ( [5] ) models, and scale models.

(2) Choose the classification of the models most related to sentences [1] to [4] from the options **A** to **C** below, and answer using the letters.

[1] A single die was rolled once to calculate the probability of rolling a five.
[2] A layout diagram of a room was created to rearrange its furnishings.
[3] The interest after five years was calculated with an interest rate of 1%.
[4] A prediction of the weather tomorrow.

**A** Static model **B** Deterministic model **C** Probabilistic model

(3) Choose the models that correspond to [1] to [3] from the options **A** to **F** below, and answer using the letters.

[1] Enlarged model [2] Full-scale model [3] Scale model

**A** Model room **B** Globe **C** Life-size dummy
**D** Plastic model **E** DNA model **F** Molecular model

## Exercise

**1** Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentence from the options **A** to **D** below, and answer using the letters.

> The process of creating models of events or phenomena is referred to as ( [1] ). Additionally, models can be broadly classified based on their characteristics into two categories: ( [2] ) models that do not undergo changes over time, and ( [3] ) models that are influenced by changes over time.

**A** Physical **B** Static **C** Dynamic **D** Modeling

(2) Choose the models that correspond to [1] to [3] from the options **A** to **F** below, and answer using the letters.

[1] Physical model [2] Diagrammatic model [3] Mathematical model

**A** Plastic model **B** Flowchart **C** Life-size dummy
**D** Route map **E** DNA model **F** Ohm's law